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SNACK N' TRACK

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ABSTRACT

Snack n’ Track is an innovative mobile health application that employs AI technology for calorie tracking practices. Users can capture photographs of their meals, enabling the application to deploy advanced computer vision algorithms alongside a sophisticated food recognition model to accurately identify diverse food items and evaluate their nutritional composition, which includes metrics such as calories, proteins, fats, and carbohydrates. This streamlined approach obviates the need for manual data entry, thereby significantly enhancing both user convenience and data accuracy.

The application is designed with a modular architecture that encompasses a cross-platform mobile interface, a scalable backend server, and an AI-driven classification engine. This cohesive system delivers real-time, precise, and user-friendly nutritional insights, empowering users to make well-informed dietary choices and adopt more healthy eating habits.

Currently operational and fully functional, the application is undergoing continuous improvements aimed at enhancing its features and overall performance. The primary goal of this initiative is to provide a practical, user-centric, and intelligent solution that leverages advanced technology to assist individuals in effectively managing their nutritional intake. Through these efforts, Snack n’ Track aspires to redefine personalized nutrition management within the contemporary digital framework.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
NCDs	Non-Communicable Diseases
UPC	Universal Product Code
CPU	Central Processing Unit
API	Application Programming Interface
DB	Database
DL	Deep Learning
GUI	Graphical User Interface
ML	Machine Learning
OCR	Optical Character Recognition
RAM	Random Access Memory
UI	User Interface
UX	User Experience
JSON	JavaScript Object Notation
SQL	Structured Query Language
CV	Computer Vision
JPEG	Joint Photographic Experts Group
PNG	Portable Network Graphics
SDK	Software Development Kit
DBMS	Database Management System
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
CLI	Command Line Interface
IDE	Integrated Development Environment
FCM	Firebase Cloud Messaging
JWT	JSON Web Token

NOMENCLATURES

C	Calories (kcal)
P	Protein content (g)
F	Fat content (g)
CH	Carbohydrate content (g)
N	Number of food items identified
A	Accuracy of prediction (%)
t_p	Processing time per image (s)
I	Input image
M	Machine learning model
L	Loss function
E	Error rate (%)

CHAPTER 1

INTRODUCTION TO SNACK N' TRACK

1 Introduction to Snack n' Track

1.1 Background

The global prevalence of diet-related health crises, such as obesity, diabetes, and cardiovascular diseases, has escalated dramatically in recent decades. These challenges are exacerbated by sedentary lifestyles, urbanization, and increased consumption of processed foods. To address this, organizations like the World Health Organization (WHO) and the United Nations have prioritized initiatives such as the Sustainable Development Goals (SDGs), which emphasize equitable access to nutritious food and improved public health outcomes.

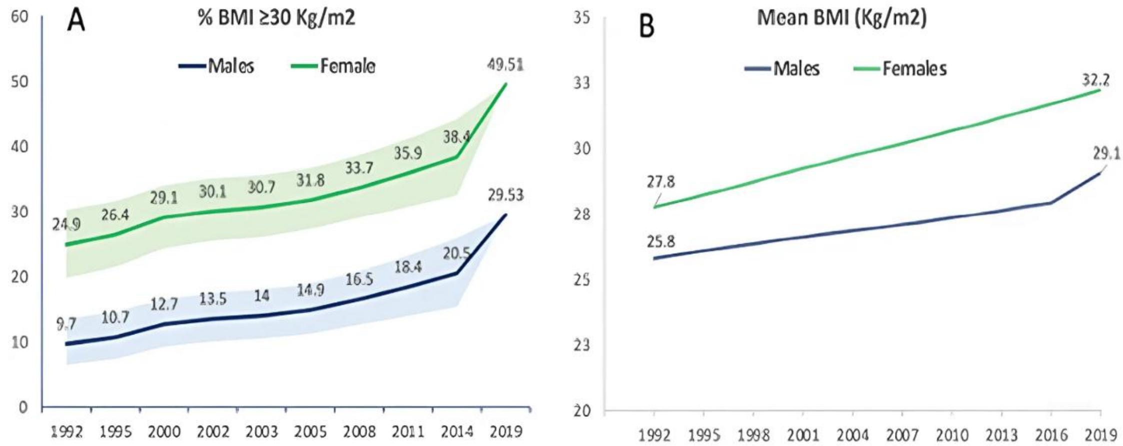

Figure 1-1 Trends in Obesity and Mean BMI Since 1992 to 2019.

Traditional methods for tracking dietary intake rely on manually logged meals, which are often time-consuming, prone to errors, and difficult to maintain over the long term. On the other hand, advancements in AI and computer vision offer new opportunities to automate nutritional analysis. Snack n' Track uses these technologies to provide an easy-to-use solution for real-time dietary monitoring. By doing so, it connects cutting-edge innovation with practical tools for managing health effectively.

1.1.1 Global Health and Nutrition Trends

The number of diet-related diseases is increasing rapidly worldwide. According to the World Health Organization, 39% of adults were overweight in 2022, and 13% were obese almost triple the obesity rates since 1975. These trends are closely linked to the growing consumption of unhealthy, high-calorie foods and inactive lifestyles. Non-communicable diseases (NCDs), such as heart disease and diabetes, now cause 71% of all deaths globally each year, with poor eating habits being a major contributing factor.

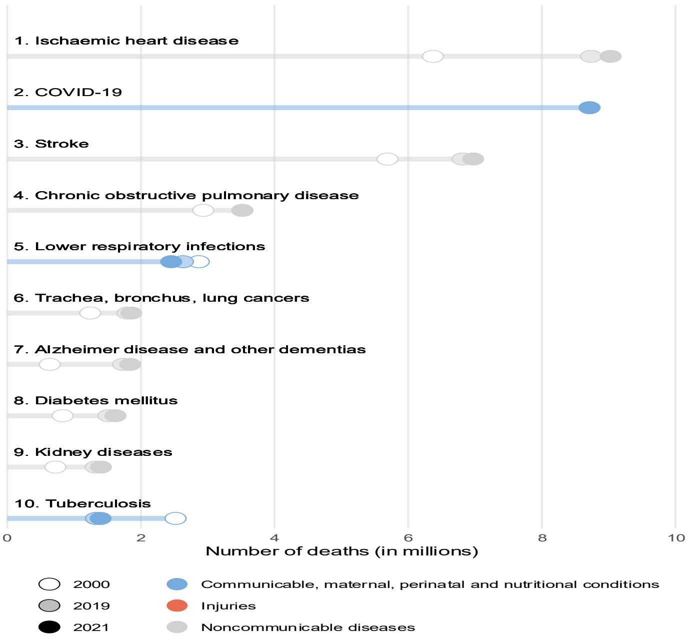

Figure 1-2 Causes of Death in 2021 Globally.

To address these issues, healthcare systems are focusing more on prevention. The WHO Global Action Plan for the Prevention and Control of NCDs highlights the importance of dietary monitoring to reduce the burden of these diseases. At the same time, personalized nutrition supported by AI and big data is gaining attention as an effective solution. For example, a 2023 study in Nature Food showed that AI-driven dietary recommendations could reduce the risk of Type 2 diabetes by 30%.

Sustainability is also becoming a key part of discussions about nutrition. The EAT-Lancet Commission promotes diets that support both human health and environmental sustainability. This highlights the need for tools that not only track nutritional value but also consider the environmental impact of food choices. These global trends highlight the importance of innovative solutions like Snack n' Track, which help users make better dietary decisions while supporting health and sustainability goals.

1.1.2 Traditional Dietary Tracking Methods

Traditional dietary tracking methods have long depended on manual input, where users log their meals by entering details such as food type, portion size, and nutritional content. Popular applications like MyFitnessPal and Lose It! provide extensive food databases for users to search and select from. While these tools have helped raise awareness about nutrition, they face several limitations that reduce their effectiveness and user engagement.

One major challenge is the time-consuming process of manual logging. Users must manually enter each food item, often estimating portion sizes or searching through large databases. This can lead to frustration and cause users to stop using the app over time. Additionally, these methods are prone to human error, as users may inaccurately estimate portion sizes or misidentify foods, resulting in unreliable nutritional data.

Another limitation is the lack of support for local and traditional foods. Many existing apps focus mainly on Western cuisines, leaving users in regions with diverse culinary traditions underserved. For example, someone eating Middle Eastern or Asian dishes may struggle to find accurate nutritional information, making these tools less useful in global contexts.

Moreover, traditional methods often fail to integrate well with modern technologies. While some apps allow barcode scanning (e.g., using UPC codes), this feature only works for packaged foods and excludes homemade or restaurant meals. The absence of advanced features like image recognition or automated analysis creates a gap between what users expect and what current solutions offer.

Despite these drawbacks, traditional dietary tracking methods have laid the groundwork for more innovative approaches. By addressing their limitations through automation and AI-driven solutions, tools like Snack n' Track aim to revolutionize how individuals monitor their nutritional intake.

1.1.3 Rise of AI and Computer Vision in Nutrition

Traditional dietary tracking methods rely on manual input, where users log meals by entering details like food type, portion size, and nutritional content. Applications such as MyFitnessPal and Lose It! offer extensive food databases, but they face significant limitations. Manual logging is time-consuming and prone to errors, as users often inaccurately estimate portions or misidentify foods, leading to unreliable data. Additionally, these tools predominantly focus on Western cuisines, leaving users with diverse culinary traditions, such as Middle Eastern or Asian dishes, underserved. The lack of support for local foods, combined with limited integration with modern technologies like image recognition or automated analysis, further reduces their effectiveness and user engagement.

Despite these challenges, traditional methods have paved the way for more innovative solutions. By addressing issues like manual input, inaccurate data, and limited cultural inclusivity, AI-driven tools like Snack n' Track aim to transform dietary tracking. Automation and advanced features, such as real-time food recognition and nutritional analysis, bridge the gap between user expectations and current capabilities, offering a more convenient and accurate way to monitor nutritional intake.

1.2 Project Overview

1.2.1 Project Concept

Snack n' Track is an innovative mobile application that transforms dietary tracking by leveraging Artificial Intelligence (AI) and Computer Vision (CV). The core idea is to allow users to effortlessly monitor their nutritional intake by simply taking photos of their meals. Using advanced algorithms, the app identifies food items and provides real-time nutritional insights, including calories, proteins, fats, and carbohydrates.

By eliminating manual data entry, a significant drawback of traditional methods Snack n' Track addresses a key user frustration. Its modular architecture ensures scalability and flexibility, enabling features like cross-platform compatibility, real-time analysis, and seamless integration with external devices such as fitness trackers.

1.2.2 System Components

Snack n' Track is built on a modular architecture that integrates three core components: the frontend, backend, and AI-driven classification engine. Each component plays a critical role in ensuring the system's functionality, scalability, and user-friendliness.

1.2.2.1 Frontend (Cross-Platform Mobile Interface):

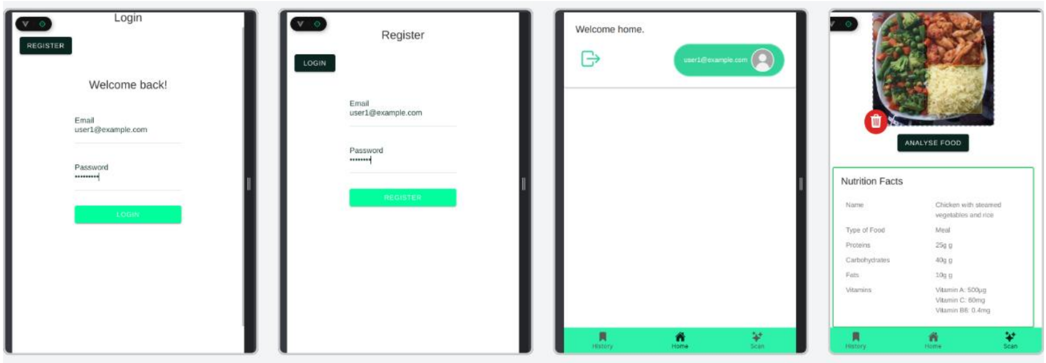

Figure 1-3 Mobile Application.

The frontend serves as the primary point of interaction for users. Developed using frameworks like Flutter, it ensures cross-platform compatibility, allowing the application to run seamlessly on both Android and iOS devices. The interface is designed to be intuitive, enabling users to capture food images, view nutritional breakdowns, and track their dietary habits effortlessly.

1.2.2.2 Backend (Scalable Server Infrastructure)

The backend acts as the backbone of the system, managing data processing, storage, and communication between the frontend and AI model. Built using technologies such as Node.js and Firebase, it supports real-time data synchronization, user authentication, and scalable database management. This ensures the system can handle increasing user loads without compromising performance.

1.2.2.3 AI-Driven Classification Engine

At the heart of Snack n' Track lies the AI-driven classification engine, which processes food images to identify items and calculate their nutritional content. Leveraging

Convolutional Neural Networks (CNNs), this component achieves high accuracy in food recognition and portion estimation [20]. The engine is trained on a diverse dataset of food images, ensuring robust performance across various cuisines and meal types.

Together, these components form a cohesive system that delivers real-time, accurate, and personalized nutritional insights, empowering users to make informed dietary decisions.

1.2.3 Key Features

This project offers a range of innovative features designed to enhance user experience, improve accuracy, and promote healthier eating habits. These features leverage the power of AI, computer vision, and cross-platform development to deliver a seamless and intelligent dietary tracking solution.

1.2.3.1 Real-Time Food Recognition

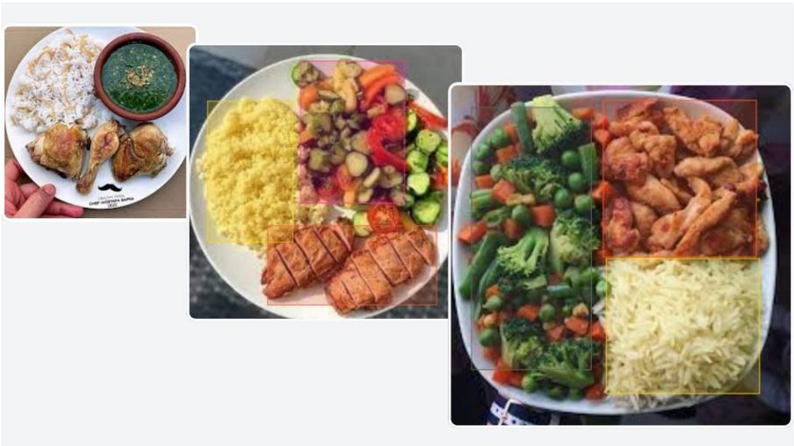

Figure 1-4 AI Model.

Users can capture images of their meals using their smartphone cameras, and the application processes these images in real-time to identify food items and provide detailed nutritional breakdowns. This feature eliminates the need for manual input, saving time and reducing errors.

1.2.3.2 Nutritional Analysis

The application calculates key nutritional metrics such as calories, proteins, fats, and carbohydrates for each meal. It also aggregates daily intake data, enabling users to monitor their progress toward health goals.

1.2.3.3 User-Friendly Interface

The intuitive UI/UX design prioritizes simplicity and ease of use, ensuring that users of all technical backgrounds can navigate the application effortlessly.

1.2.4 Target Users and Use Cases

Snack n’ Track is designed to cater to a diverse range of users, addressing various needs and scenarios in dietary tracking. The application's versatility makes it suitable for individuals, healthcare professionals, and organizations focused on wellness and nutrition.

1.2.4.1 Health-Conscious Individuals

The primary target audience includes individuals seeking to monitor their daily nutritional intake to achieve personal health goals, such as weight management, muscle gain, or improved overall well-being. By providing real-time nutritional insights, Snack n’ Track empowers users to make informed dietary decisions.

1.2.4.2 Fitness Enthusiasts

Athletes and fitness enthusiasts can leverage the app to track macronutrient intake (e.g., proteins, fats, carbohydrates) and optimize their diets for performance and recovery. Integration with external devices like fitness trackers enhances its utility by combining activity data with nutritional analysis.

1.2.4.3 Global Users with Diverse Diets

Unlike traditional apps that focus primarily on Western cuisines, Snack n’ Track supports a wide variety of local and traditional foods. This makes it an ideal solution for users in regions with diverse culinary traditions, ensuring accurate nutritional information for dishes like Egyptian or Middle Eastern meals.

1.2.5 Project Status and Development Stage

Snack n' Track is currently operational and fully functional. The application has transitioned from the initial conceptualization and design phases to an implementation stage, where it successfully integrates AI-driven food recognition, real-time nutritional analysis, and cross-platform compatibility.

Despite its current functionality, Snack n' Track remains an evolving project. Continuous improvements are being made to enhance the AI model's accuracy, expand the database of supported foods, and refine the user experience.

1.3 Motivation

1.3.1 Why We Chose This Problem

The decision to address dietary tracking stems from the growing global health crisis caused by poor nutrition, including rising rates of obesity, diabetes, and cardiovascular diseases. Traditional dietary tracking methods, such as manual logging, are tedious and error-prone, leading to low user engagement and unreliable data. Additionally, existing AI-based solutions often lack accuracy in portion estimation, support for diverse cuisines, and personalization, leaving significant gaps in usability and inclusivity.

Snack n' Track was developed to overcome these challenges by providing an automated, intelligent, and user-friendly solution. By leveraging AI and computer vision, the application eliminates the need for manual input, offering real-time nutritional insights that empower users to make informed dietary choices. This innovation not only enhances convenience and accuracy but also addresses the limitations of current tools, making dietary tracking accessible and engaging for a global audience.

1.3.2 Technological Opportunity

The advancements in AI, computer vision, and cloud computing provide a unique opportunity to transform dietary tracking. Snack n' Track leverages convolutional neural networks (CNNs) for precise food recognition, eliminating the need for manual logging and supporting diverse cuisines, including Egyptian dishes. Cloud-based solutions like

Firebase ensure scalability, real-time data synchronization, and reliability, even when the AI model is unavailable, thanks to a fallback system using ChatGPT API.

The application avoids frameworks like Flutter, focusing instead on a minimalist frontend to prioritize backend robustness. This ensures compatibility across devices while delivering core functionalities such as image capture, nutritional feedback, and progress tracking. Integration with platforms like Fitbit, Apple Health, and Google Fit enhances its value by combining dietary data with activity metrics.

1.3.3 Personal and Societal Impact

Snack n' Track has the potential to create significant personal and societal benefits by promoting healthier eating habits and addressing global health challenges. On a personal level, the application empowers users to track their meals accurately, providing real-time nutritional insights and personalized recommendations to help them achieve fitness and health goals. This is particularly valuable for individuals managing chronic conditions like diabetes or obesity, as well as athletes and fitness enthusiasts seeking precise dietary control.

On a societal level, Snack n' Track addresses the growing public health crisis caused by poor nutrition, such as rising rates of obesity and cardiovascular diseases. By supporting diverse cuisines, including Egyptian foods, it ensures inclusivity and accessibility for regional populations often overlooked by existing solutions. Its integration with health platforms like Fitbit and Apple Health fosters holistic well-being, combining dietary data with activity metrics. This not only reduces the burden on healthcare systems but also aligns with global trends toward preventive healthcare, ultimately contributing to healthier communities and improved quality of life.

1.3.4 Alignment with Global Trends

Snack n' Track aligns seamlessly with global trends in health, technology, and inclusivity by addressing the growing demand for personalized, data-driven wellness solutions. As awareness of nutrition's role in preventing chronic diseases like obesity, diabetes, and cardiovascular conditions continues to rise, there is an increasing need for tools that simplify dietary tracking and promote healthier lifestyles. By leveraging AI and computer vision, Snack n’ Track offers precise, real-time nutritional insights, catering to diverse populations, including regional cuisines like Egyptian foods, which are often overlooked by existing solutions. Its integration with wearable devices and health platforms reflects the broader shift toward interconnected health ecosystems, while its focus on accessibility and scalability positions it at the forefront of digital health innovation, supporting global movements toward preventive healthcare and sustainable wellness.

1.4 Problem Statement

1.4.1 Existing Challenges in Dietary Tracking

Traditional dietary tracking methods face significant challenges that limit their effectiveness and user engagement. Manual logging, the most common approach, is time-consuming and prone to human error, as users must estimate portion sizes or search through extensive databases, often leading to inaccurate data [1]. Additionally, many existing applications lack support for diverse cuisines, particularly regional and traditional foods, leaving users in areas like the Middle East or Asia underserved [2]. Even AI-based solutions, while promising, encounter limitations such as inadequate accuracy in food recognition, especially for mixed dishes, and insufficient personalization for individual dietary needs [3]. Furthermore, integration with external devices like fitness trackers remains underdeveloped, creating gaps in holistic health monitoring. These challenges highlight the need for a more innovative, inclusive, and accurate solution to address the shortcomings of current dietary tracking systems.

1.4.2 Limitations in AI-based Alternatives

While AI-based dietary tracking solutions have shown promise, they still face significant limitations that hinder their widespread adoption and effectiveness. One major challenge is the lack of accuracy in recognizing diverse and complex meals, particularly mixed dishes or culturally specific cuisines like Egyptian foods, due to insufficient training datasets. Additionally, many AI models struggle with portion estimation, which can lead to inaccurate nutritional insights and undermine user trust. Another limitation lies in the computational demands of these systems; real-time processing often requires robust hardware or cloud connectivity, which may not always be feasible for users in areas with limited internet access. Furthermore, existing AI-driven applications often fail to provide seamless integration with external devices like fitness trackers, creating gaps in holistic health monitoring. These shortcomings highlight the need for more inclusive, accurate, and adaptable AI solutions that address both technical and user-centric challenges.

1.4.3 Project Statement

Snack n' Track addresses the existing gaps in dietary tracking by leveraging AI and computer vision to provide an innovative, user-friendly solution for nutritional management. The project aims to overcome the limitations of traditional methods, such as manual logging and inaccurate portion estimation, while also tackling challenges faced by current AI-based alternatives, including the lack of support for diverse cuisines like Egyptian foods and inconsistent model accuracy. By combining advanced food recognition algorithms with a seamless mobile application, Snack n' Track enables users to effortlessly track their meals, receive precise nutritional insights, and integrate health metrics from wearable devices. This initiative not only simplifies dietary monitoring but also empowers individuals to make informed decisions about their health, ultimately contributing to improved wellness outcomes and aligning with global trends toward preventive healthcare.

The project focuses on creating a scalable, modular system that integrates a cross-platform mobile interface, a robust backend infrastructure, and an AI-driven classification engine. Through continuous improvements and user-centric design, Snack n' Track strives to redefine personalized nutrition management by offering a practical, accurate, and inclusive tool tailored to the needs of diverse users—from health-conscious individuals to those managing chronic conditions. By addressing technical and usability challenges, the project seeks to establish itself as a leading solution in the digital health ecosystem.

1.4.4 Objectives of the Project

The main goal of Snack n’ Track is to provide a simple, accurate, and user-friendly solution for dietary tracking by using AI and computer vision technologies. The project aims to solve the problems of traditional methods, such as manual logging and human errors, while also addressing the limitations of existing AI-based tools. These limitations include poor recognition of diverse cuisines, inaccurate portion estimation, and lack of personalized insights. By developing a system that can identify food items from images and calculate their nutritional values, Snack n’ Track seeks to make dietary tracking easier and more accessible for everyone.

Another key objective is to create a scalable and modular system that combines a mobile application, a reliable backend infrastructure, and an AI-driven classification engine. This system will allow users to track their meals, monitor their health goals, and integrate data from fitness devices like Fitbit and Apple Health. The project also focuses on inclusivity by supporting regional foods, such as Egyptian cuisine, which are often ignored in global datasets. By achieving these objectives, Snack n’ Track aims to empower users to make informed decisions about their nutrition and improve their overall well-being.

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APPENDICES

APPENDICES

Appendix A.

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Appendix B.

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